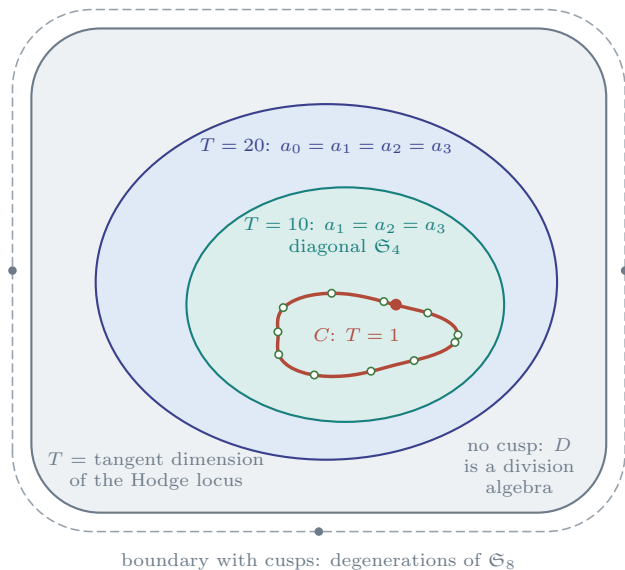


$\mathfrak{S}_8$, dim 36: principally polarised 8-folds near $X_c \times X_c$



	weight μ_1 of $\mathfrak{sl}_2(V_1)$		
	-2	0	+2
-2	3 $P_1 P_2 Q_1$	4 $P_1 P_2 Q_1 Q_3$	3 $P_1 P_2 Q_1$
0	4 $P_1 P_2 Q_1 Q_2$	8 = 7 + 1 $R P_1 P_2$ $Q_1 Q_2 Q_3 Q_4$	4 $P_1 P_2 Q_1 Q_2$
+2	3 $P_1 P_2 Q_1$	4 $P_1 P_2 Q_1 Q_3$	3 $P_1 P_2 Q_1$

$$\mathfrak{sp}^{-1,1}: 3 + 4 + 3 + 4 + 8 + 4 + 3 + 4 + 3 = 36$$

in $(0, 0)$ one kernel vector, the curve direction κ

$$HT^2 = 28 + 64 + 28: 46 \text{ blocks, kernel } \mathbb{C}\kappa \text{ if } a_0 + \text{tr } \alpha \neq 0$$

$$2P_1 = \sum_{i < j} (a_i - a_j)^2, \quad 2P_2 = \sum_{i < j} (a_i + a_j)^2, \quad R = a_2^2 + a_3^2, \quad Q_4 = (3a_1 - a_0 - a_2 - a_3)^2 + 16(a_2 - a_3)^2$$

$$33Q_1 = 3(11a_1 - a_0 - 5a_2 - 5a_3)^2 + 8(6a_2 - a_0 - 5a_3)^2 + 88(a_3 - a_0)^2, \quad S_1 = (a_0 - a_2)^2 + 3(a_1 - a_3)^2$$

a rational exceptional class has $(a_1, a_2, a_3) = (\sigma_1 \alpha, \sigma_2 \alpha, \sigma_3 \alpha)$ with $\alpha \in F \setminus \mathbb{Q}$: pairwise distinct, so every block has full rank